STA 2002

# Lecture 4

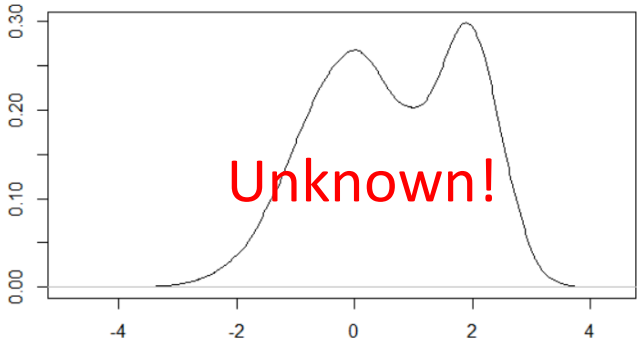
Maximum likelihood estimator (cont'd)

Method of moments

Unbiased estimator

# Estimation

Population




Population parameters $\theta$
(unknown)

Good sampling scheme?

**i.i.d. assumed in STA2002**

Sample

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| -1.28 | 1.81 | 1.45 | 0.05 | 2.08 | -1.15 | -0.34 | 2.23 | -0.20 | 1.20 |
| 1.04 | -1.12 | 1.88 | -0.86 | 0.25 | 2.39 | -0.76 | 0.64 | 2.54 | 0.54 |
| 1.67 | 1.91 | 1.72 | 2.17 | 0.15 | -1.24 | 2.42 | -0.53 | -1.36 | -0.56 |

Some sample statistic

Estimated parameters $\hat{\theta}$

Is $\hat{\theta}$ a good estimate of $\theta$ ?

## Estimator

Repeat an experiment $n$ times independently to observe the sample

$$X_1, X_2, \cdots, X_n,$$

and estimate the parameter $\theta \in \Omega$ by using observations

$$x_1, x_2, \cdots, x_n.$$

The statistics $U = u(X_1, X_2, \cdots, X_n)$ to estimate $\theta$ is called an **estimator** of $\theta$

Hope the realized estimate $\mathrm{u} = u(x_1, x_2, \cdots, x_n)$ is close to $\theta$.

Estimators are not unique → How to evaluate different estimators?

## Clarification

The use of *X* or *x*

By convention, "*x*" represents a sample, "*X*" represents a Random Variable

"***x***" represents a multi-dimensional sample, "***X***" represents a Random Vector

In the discussion of MLE, we see some books use "$X_i$", and some others use "$x_i$"

Using "*x*" stresses the estimation is based on a sample

Using "*X*" stresses the procedure works for **any** sample – back to the sense of random variable

Both ways can be seen in books written by reputable statisticians. Both are correct.

# Maximum likelihood

The **maximum likelihood estimator (MLE)** of θ is that value of θ that maximizes the likelihood— makes the observed data "most probable" or "most likely."

If the $X_i$ are assumed to be i.i.d., $$L(\theta) = \prod_{i=1}^{n} f(X_i|\theta)$$

Maximizing log-likelihood is equivalent to maximizing likelihood, since the logarithm is a monotonic function.

i.i.d. sample, the **log likelihood** is

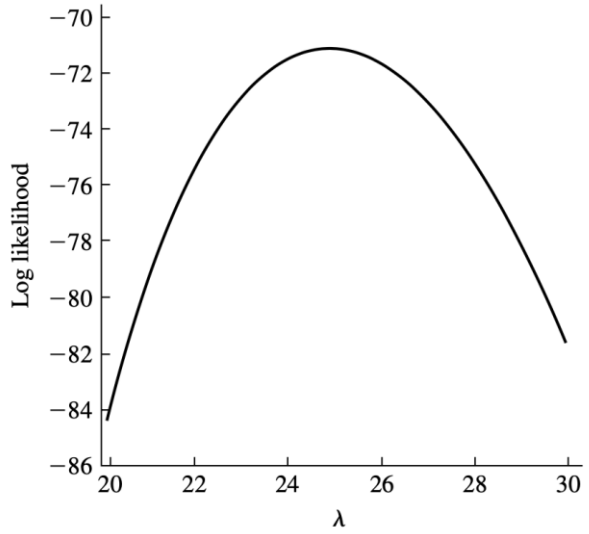
$$l(\theta) = \sum_{i=1}^{n} \log[f(X_i|\theta)]$$

# Maximum Likelihood

The **maximum likelihood estimate (MLE)** of θ is obtained by solving

$$l'(\theta) = 0$$

and checking $l''(\theta) < 0$ when evaluated at $\hat{\theta}$ (sometimes omitted when there is a single root for $l'(\theta) = 0$).

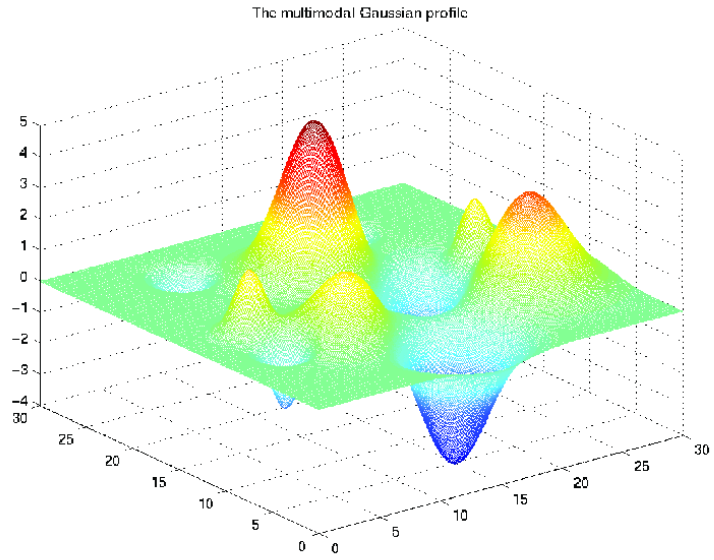



When the likelihood is more complicated,

$$l'(\theta) = 0$$

$$l''(\theta) < 0$$

only guarantees a local optimum. May need to compare different L({$x_i$}; $\widehat{\theta_k}$} to find the best among multiple roots. But usually analytical solution is not even available for such comparisons.




(Not our concern in this course)

# Maximum likelihood

Exponential(1/θ)

$$f(x;\theta) = \frac{1}{\theta} e^{-x/\theta}$$

$$L(\theta \mid x_1, \ldots, x_n) = \frac{1}{\theta^n} \exp\left(-\frac{n\bar{x}}{\theta}\right)$$

$$l(\theta \mid x_1, \ldots, x_n) = -n \ln\theta - \frac{n\bar{x}}{\theta}$$

$$-\frac{n}{\theta} + \frac{n\bar{x}}{\theta^2} = 0$$

$$\hat{\theta} = \bar{x}$$

# Maximum likelihood

Geometric distribution $f(x;p) = (1-p)^{x-1}p$

$$L(p) = (1-p)^{x_1-1}p(1-p)^{x_2-1}p\cdots(1-p)^{x_n-1}p$$

$$= p^n(1-p)^{\Sigma x_i - n}$$

$$\ln L(p) = n\ln p + \left(\sum_{i=1}^{n} x_i - n\right)\ln(1-p)$$

$$\frac{d\ln L(p)}{dp} = \frac{n}{p} - \frac{\sum_{i=1}^{n} x_i - n}{1-p} = 0$$

$$p = \frac{n}{\sum_{i=1}^{n} x_i} = \frac{1}{\bar{x}}$$

# Maximum likelihood

Uniform distribution

$$f(x) = \begin{cases} \frac{1}{R-L} & L \le x \le R \\ 0 & \text{otherwise.} \end{cases}$$

$$\max L(R, L) = \max \left( \frac{1}{R-L} \right)^n$$
$$subject\ to\ R \ge \max(x)\,, L \le \min(x)$$

It is equivalent to:

$$\min\ (R-L)$$
$$subject\ to\ R \ge \max(x)\,, L \le \min(x)$$

Solution is $R = \max(x)\,, L = \min(x)$

## Maximum likelihood: No-closed form

$$X_i \; i.i.d \sim Gamma(\alpha, \lambda)$$

$$f(x|\alpha, \lambda) = \frac{1}{\Gamma(\alpha)}\lambda^{\alpha}x^{\alpha-1}e^{-\lambda x}, \qquad 0 \leq x < \infty$$

$$l(\alpha, \lambda) = \sum_{i=1}^{n}[\alpha \log \lambda + (\alpha - 1)\log X_i - \lambda X_i - \log \Gamma(\alpha)]$$

$$= n\alpha \log \lambda + (\alpha - 1)\sum_{i=1}^{n} \log X_i - \lambda \sum_{i=1}^{n} X_i - n \log \Gamma(\alpha)$$

Setting derivative to zero, can't find a closed form solution. Need to resort to numeric approach to solve for the parameters.

$$\frac{\partial l}{\partial \alpha} = n \log \lambda + \sum_{i=1}^{n} \log X_i - n\frac{\Gamma'(\alpha)}{\Gamma(\alpha)}$$

$$\frac{\partial l}{\partial \lambda} = \frac{n\alpha}{\lambda} - \sum_{i=1}^{n} X_i$$

No need to go into details

# Method of moments

The $k$th moment of a probability law is defined as

$$\mu_k = E(X^k)$$

Parameters of the distribution $\theta_1, \theta_2, \ldots$ can be expressed as

$$\theta_1 = f(\mu_1, \mu_2, \ldots)$$
$$\theta_2 = g(\mu_1, \mu_2, \ldots)$$
$$\ldots \ldots$$

When we have a sample, moments of the distribution are easily estimated using sample moments.

the $k$th **sample moment** is defined as

$$\hat{\mu}_k = \frac{1}{n}\sum_{i=1}^{n} X_i^k$$

$$\hat{\theta}_1 = f(\hat{\mu}_1, \hat{\mu}_2, \ldots), \hat{\theta}_2 = g(\hat{\mu}_1, \hat{\mu}_2, \ldots), \ldots$$

We say $\hat{\theta}_1, \hat{\theta}_2, \ldots$ are estimated by the method of moments, or called **method of moments estimators (MME)**

# Method of moments

Gamma distribution:

$$f(x|\alpha, \lambda) = \frac{1}{\Gamma(\alpha)}\lambda^{\alpha}x^{\alpha-1}e^{-\lambda x}, \qquad 0 \le x < \infty$$

Its MGF:

$$M(t) = \int_0^{\infty} e^{tx}\frac{\lambda^{\alpha}}{\Gamma(\alpha)}x^{\alpha-1}e^{-\lambda x}\,dx$$

$$= \frac{\lambda^{\alpha}}{\Gamma(\alpha)}\int_0^{\infty} x^{\alpha-1}e^{x(t-\lambda)}\,dx$$

The latter integral converges for $t < \lambda$ and can be evaluated by relating it to the gamma density having parameters $\alpha$ and $\lambda - t$. We thus obtain

$$M(t) = \frac{\lambda^{\alpha}}{\Gamma(\alpha)}\left(\frac{\Gamma(\alpha)}{(\lambda - t)^{\alpha}}\right) = \left(\frac{\lambda}{\lambda - t}\right)^{\alpha}$$

# Method of moments

Take derivative against t, evaluate at t=0, to find the first two moments:

$$\mu_1 = \frac{\alpha}{\lambda}$$

$$\mu_2 = \frac{\alpha(\alpha+1)}{\lambda^2}$$

Solve for parameters when moments are estimated from sample:

$$\mu_2 = \mu_1^2 + \frac{\mu_1}{\lambda}$$

$$\lambda = \frac{\mu_1}{\mu_2 - \mu_1^2}$$

$$\alpha = \lambda\mu_1 = \frac{\mu_1^2}{\mu_2 - \mu_1^2}$$

# Method of moments

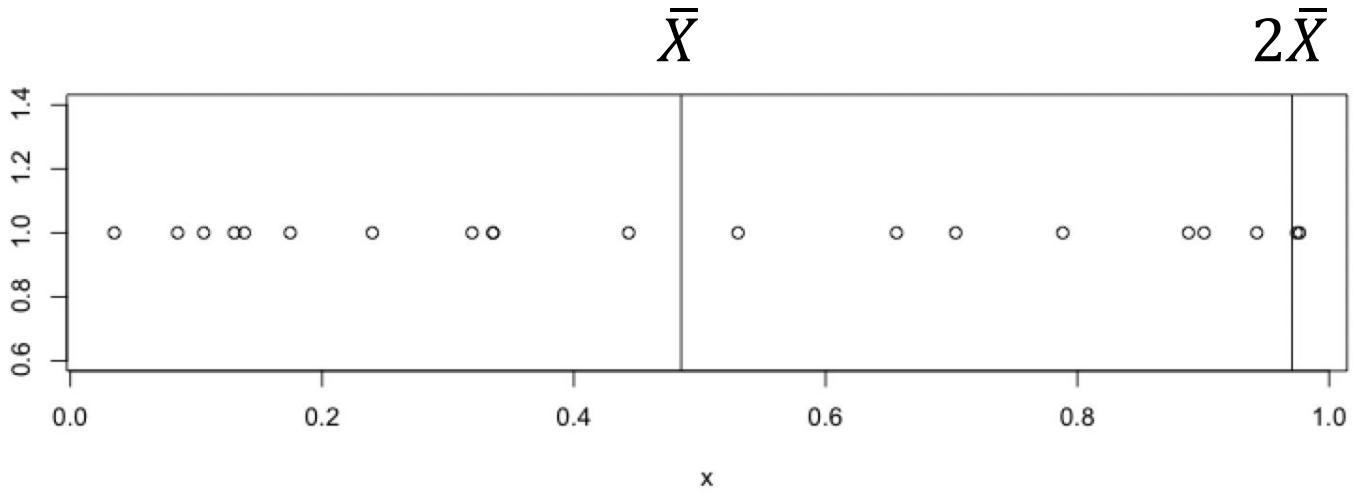
Uniform distribution:

First moment: $$\int_L^R \frac{X}{R-L} dX = \frac{1}{R-L}\int_L^R X dX = \frac{1}{R-L}\int_L^R d\left(\frac{X^2}{2}\right) = \frac{R+L}{2}$$

For a simple case, suppose L=0 is given, and $x_i \geq 0, \forall i$

Method of moment result: $\frac{R}{2} = \bar{X}$

But this can sometimes happen:



# Method of moments

Poisson distribution

$$m_Z(s) = \exp\left(\lambda(e^s - 1)\right)$$

$$E(X) = m'_X(0) = \lambda e^0 \exp\left(\lambda(e^0 - 1)\right) = \lambda$$

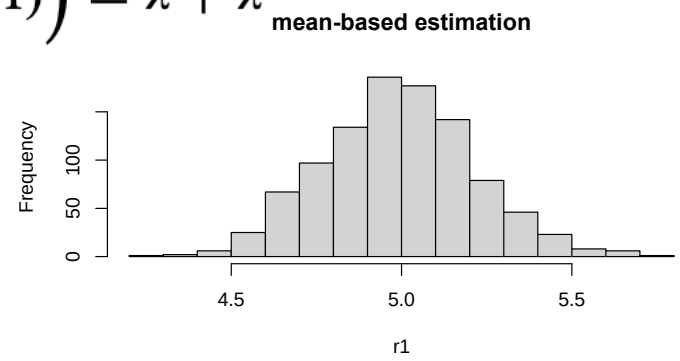
$$E(X^2) = m''_X(0) = \lambda e^0 \exp\left(\lambda(e^0 - 1)\right) + \left(\lambda e^0\right)^2 \exp\left(\lambda(e^0 - 1)\right) = \lambda + \lambda^2$$

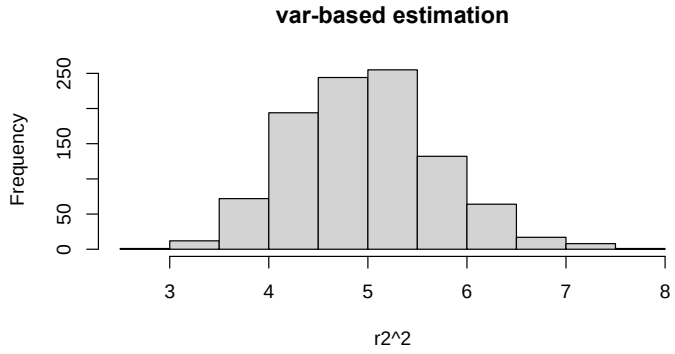
$$\hat{\mu}_1 = \overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \qquad \hat{\mu}_2 = \frac{1}{n}\sum X_i^2$$

$$\hat{\lambda}_1 = \hat{\mu}_1 \qquad \hat{\lambda}_2 = \hat{\mu}_2 - (\hat{\mu}_1)^2$$

- Why not use variance?

It is not as “stable”. “Not stable”– higher variation



# Method of moments

Let $X \sim \text{Exponential}(\lambda)$

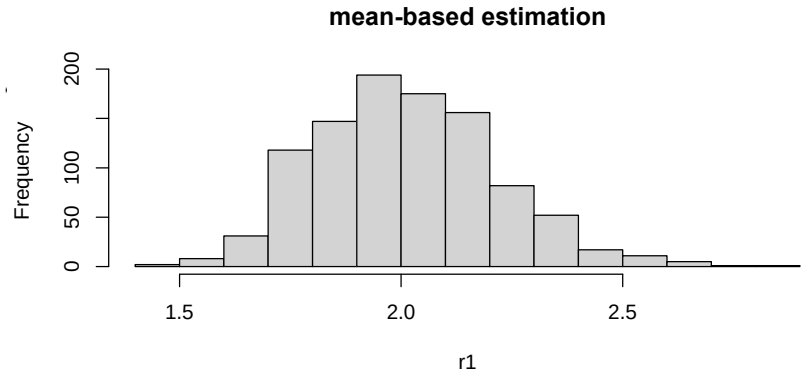
$$m_X(s) = \lambda(\lambda - s)^{-1}$$

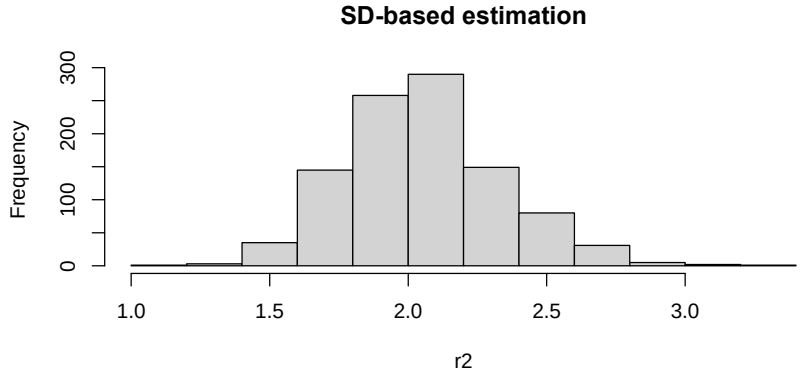
$$E(X) = m'_X(0) = \lambda(\lambda - 0)^{-2} = \lambda/\lambda^2 = 1/\lambda$$

$$\text{Var}(X) = E(X^2) - (E(X))^2 = (2/\lambda^2) - (1/\lambda)^2 = 1/\lambda^2$$

$$\hat{\lambda}_1 = \frac{1}{\bar{X}}, \qquad \hat{\lambda}_2 = \frac{1}{\sqrt{\frac{1}{n}\sum X_i^2 - \bar{X}^2}}$$

When there are more than one parameter to estimate, calculate as much lower moment as possible to solve for the parameters.



# MME and MLE

Usually MLE is preferred if available

When large amount of computation is needed, MME can be useful due to speed

Sometimes MME can give analytical solution while MLE cannot;  in this case MLE can still use numerical solution (Ex: gamma distribution)

Sometimes MME gives unrealistic solution (Ex: uniform distribution)

# Sampling distribution of the estimates

Our knowledge linking sample mean and variance estimates:

$$E\left(\frac{X_1 + X_2 + \cdots + X_n}{n}\right) = \frac{E(X_1) + E(X_2) + \cdots + E(X_n)}{n} = \mu$$

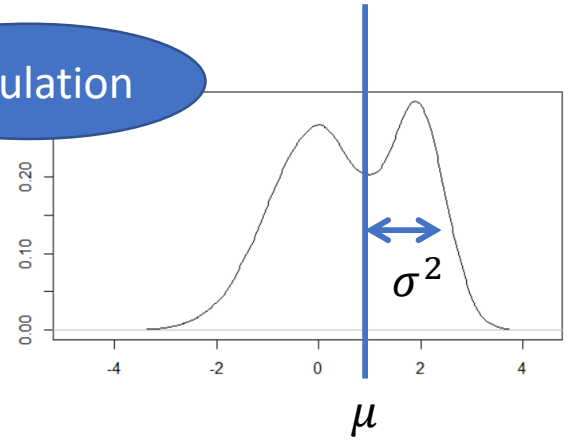
$$var\left(\frac{X_1 + X_2 + \cdots + X_n}{n}\right) = \frac{var(X_1) + var(X_2) + \cdots + var(X_n)}{n^2} = \frac{\sigma^2}{n}$$

CLT says: As $n \to \infty, \bar{X} \sim N(\mu, \frac{\sigma^2}{n})$

Sample mean $\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$

is a sample from

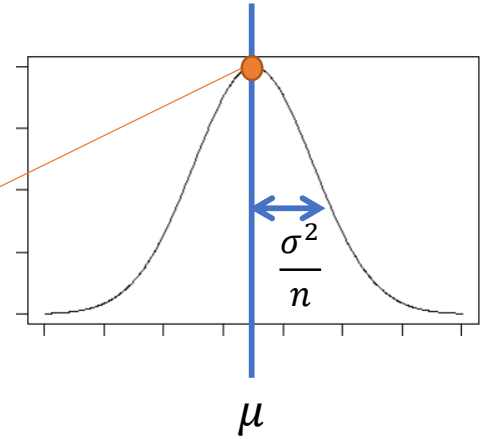
Population




Sample

-0.01 0.01 -0.70 -0.71 -0.78
-1.33 -0.27 -1.36 -1.26 -0.62

What $\mu$ can you get here?



## Sampling distribution of the estimates

### The Poisson example

Back to MGF:

PROPERTY **D**

If $X$ and $Y$ are independent random variables with mgf's $M_X$ and $M_Y$ and $Z = X + Y$, then $M_Z(t) = M_X(t)M_Y(t)$ on the common interval where both mgf's exist.

For two Poisson RVs,

$$e^{\lambda(e^t-1)}e^{\mu(e^t-1)} = e^{(\lambda+\mu)(e^t-1)}$$

$X + Y$ is Poisson with parameter $\lambda + \mu$

# Sampling distribution of the estimates

## The Poisson example

The sum of n Poisson($\lambda_0$) samples, S, is one sample from Poisson($n\lambda_0$).

$$\hat{\lambda} = \overline{X}$$

$$E(\hat{\lambda}) = \frac{1}{n}E(S) = \lambda_0$$

The estimate is **unbiased:** the sampling distribution is centered at $\lambda_0$.

$$\mathrm{Var}(\hat{\lambda}) = \frac{1}{n^2}\mathrm{Var}(S) = \frac{\lambda_0}{n}$$

The sampling distribution becomes more concentrated about $\lambda_0$ as *n* increases. The standard deviation of this distribution is called the **standard error** of $\hat{\lambda}$:

$$\sigma_{\hat{\lambda}} = \sqrt{\frac{\lambda_0}{n}}$$

The **estimated standard error**

$$s_{\hat{\lambda}} = \sqrt{\frac{\hat{\lambda}}{n}}$$

# Sampling distribution of the estimates

Ex:

X:

| | | | | | |
|---|---|---|---|---|---|
| 31 | 29 | 19 | 18 | 31 | 28 |
| 34 | 27 | 34 | 30 | 16 | 18 |
| 26 | 27 | 27 | 18 | 24 | 22 |
| 28 | 24 | 21 | 17 | 24 | |

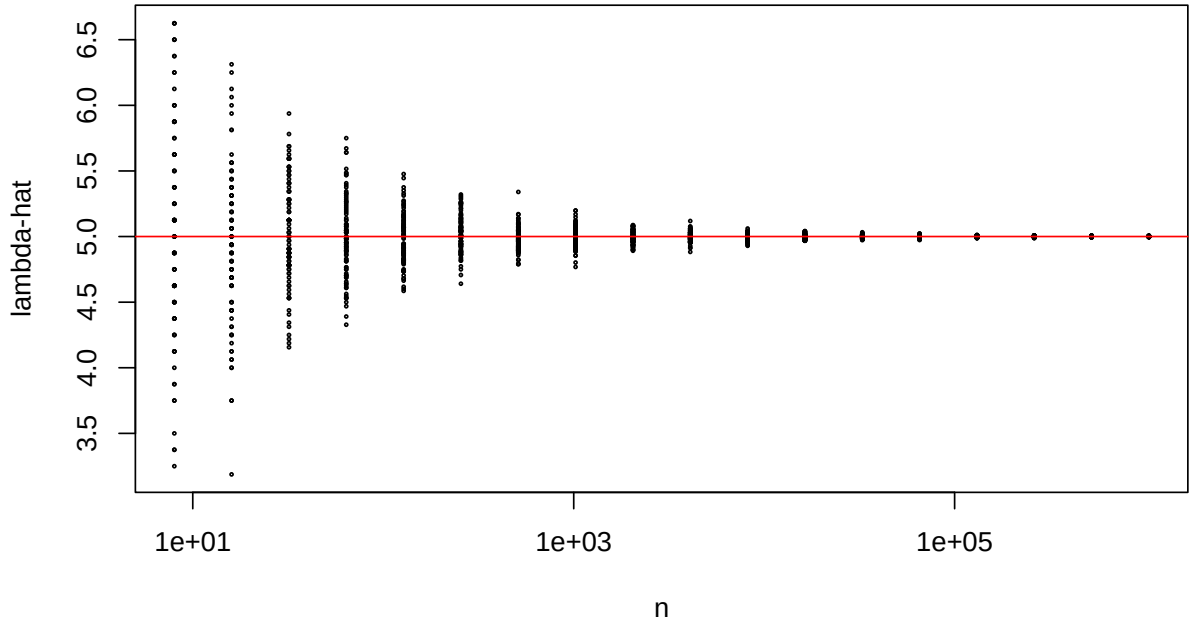
$$\hat{\lambda} = \overline{X} = 24.9$$

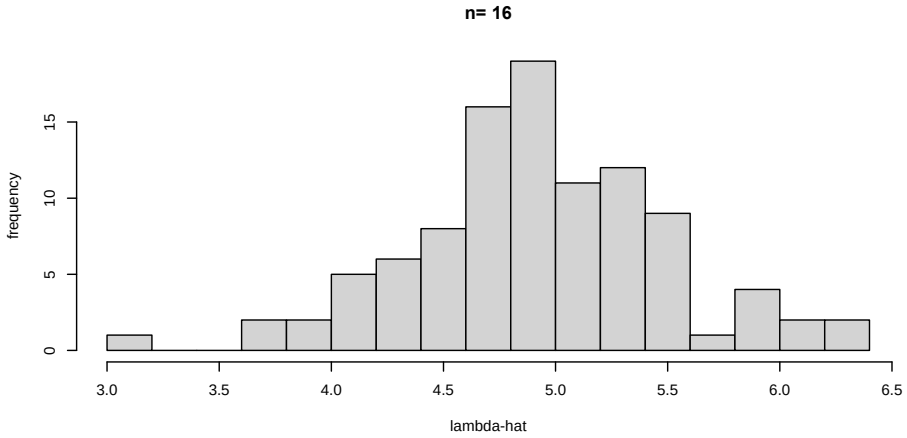
$$s_{\hat{\lambda}} = \sqrt{\frac{24.9}{23}} = 1.04$$

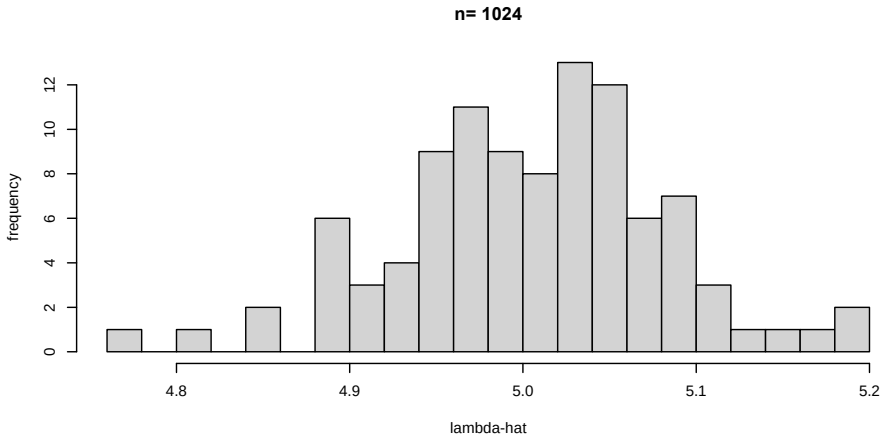
The sampling distribution of $\hat{\lambda}$ is approximately normal, centered at the true value $\lambda_0$ with standard deviation 1.04.

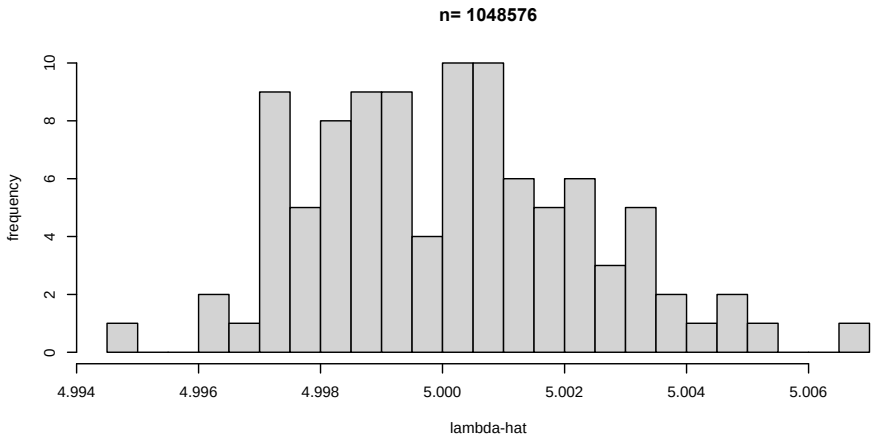
# Sampling distribution of the estimates

## The Poisson example



## Unbiased estimator

How to evaluate an estimator $\hat{\theta}$ of $\theta$?

If $\mathrm{E}\big(u(X_1, X_2, \cdots, X_n)\big) = \theta$, then the statistics $u(X_1, X_2, \cdots, X_n)$ is called an unbiased estimator of $\theta$. Otherwise, it is said to be biased.

Ex:
MLE or MME for i.i.d. Bernoulli RVs

$$E(\hat{p}) = E(\bar{X}) = p$$

MLE or MME for i.i.d. Poisson RVs

$$E\big(\hat{\lambda}\big) = E(\bar{X}) = \lambda$$

# Unbiased estimator

Exponential(1/θ)

$$\hat{\theta} = \bar{x}$$

$$E(\hat{\theta}) = E(\bar{X}) = \theta$$

$\hat{\theta}$ is unbiased.

Normal distribution

$$\hat{\mu} = \overline{X}$$

$$\hat{\sigma} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2}$$

$$E(\hat{\mu}) = E(\bar{X}) = \mu$$

$\hat{\mu}$ is unbiased.

# Unbiased estimator

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$$

The distribution of $(n-1)S^2/\sigma^2$ is the chi-square distribution with $n-1$ degrees of freedom.

$$E(S^2) = E\left[\frac{\sigma^2}{n-1}\frac{(n-1)S^2}{\sigma^2}\right] = \frac{\sigma^2}{n-1}(n-1) = \sigma^2$$

Then $\hat{\sigma} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2}$ must be biased.

## How to evaluate different estimators?

- Unbiased: An estimator with the expectation equal to $\theta$
- An estimator with a smaller variance
- Analytical from
- Avoid unrealistic solutions